Day 1

SPARC Workshop Cum International Symposium on Neuroanatomy and Diffusion MR Imaging based biomarker for TBI
December 05-07, 2023
Organizers : Prof. BV Rathish Kumar, Prof. Walt Schneider, Prof. S.K. Pathak

- Registration (9:00 AM – 9:30 AM)
- Inaugural (9:30 AM – 9:45 AM)
- Tea (9:45 AM – 10:00 AM)
- Lecture by **Walt** on TBI for this conference (45 + 5 minutes) (10:00 – 11:00)
- Introduction to Neuroanatomy (60 minutes) – **Sudhir (11:12)**
  - Highlighting white matter tracts.
  - Essential features of the Cortex and subcortical regions.
  - Identifying features in diffusion MRI for tract location.
  - Introduction to software: FREESURFER, FSL.
  - Demo and exercises for cortical area identification.
- Tea 12:00 – 12:15
- Fundamental MRI principles (60 minutes)- **Durgesh (12:15-13:15)**
  - Lamor Frequency,
  - Resonance,
  - Tissue contrast T1, T2,
  - Image formation k-space

  LUNCH 13:15 – 14:30
- Introduction to Diffusion MR Imaging (60 minutes) – **Sudhir 14:30 – 16:00**
  - Basis of Diffusion MRI
  - Mathematical Modeling of Diffusion Signal
  - Imaging Protocol used for Model.
  - Parametric and non-Parametric models (introduction)

  TEA BREAK 16:00 – 16:15

  Preprocessing of DMRI /Demos (16:15 – 17:30)

  Day 1 Break

Day 2

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- **Talk by Walt** on Validation using Phantom and ex-vivo imaging (60 minutes) 10:00 – 11:00
- Microstructural Imaging (60 minutes) – **Sudhir (11:00 – 12:00)**
  - Basis of diffusion process in biological tissue.
  - Mathematical modeling of diffusion signal (parametric)
  - NODDI/SMT/Kurtosis Imaging
  - Tea Break: 12:00 – 12:15
  - **Sudhir (12:15- 13:15)**
  - Imaging Protocol.
  - Signal on Sphere, fiber ODF, Deconvolution of the sphere
  - Tractometry (if time permits)
  - **LUNCH 13:15 – 14:30**
- Clinical Talks (2 hour) (**Walt, Sudhir,) (14:30 – 16:00)**
  - On neurosurgery
  - TBI
  - Neurodegenerative diseases
  - **Tea Break (16:00 – 16:15)**
- Deep Learning Models(2 hour)
- **(Aditya, Rathish, Ranjeet) 16:15- 18:15)**
  - Deep Learning in medical Imaging
  - Physics-based Neural Network Models
  - Neurodegenerative diseases
- Demo (one hours) **(Sudhir, Durgesh)**
- **SPECIAL DINNER (VH) 7:30 PM.**